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REVIEW



# Types, purposes, and effectiveness of state-of-the-art technologies for second and foreign language learning

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## ABSTRACT

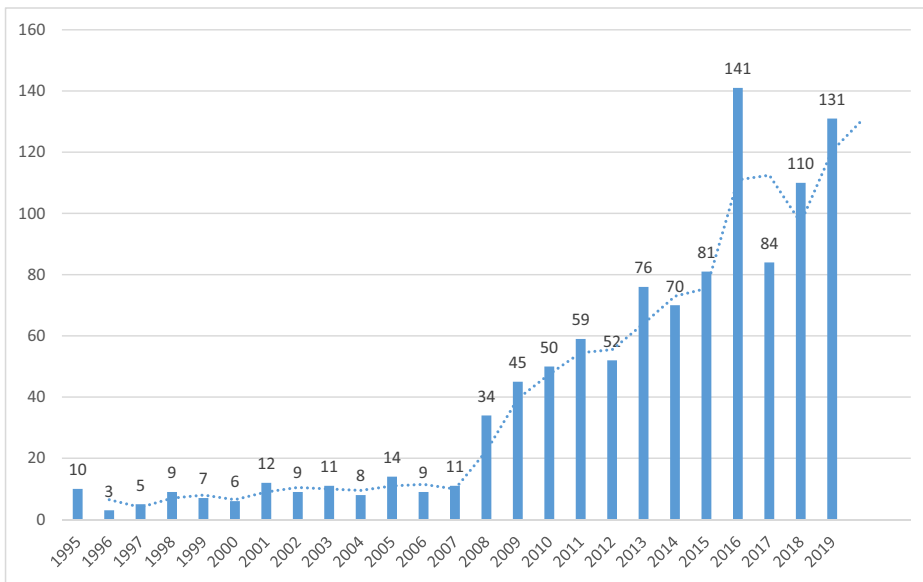
Digital technologies have been widely used to enhance language learning, the effectiveness of which has been acknowledged in the literature. With the rapid development of digital devices and technologies, increasing technologies have been used in the most recent several years, leading to more diversified approaches to language education. This exceptional advance over the past few years calls for a summary of state-of-the-art technologies that have been used to enhance language learning and promote effective learning. The present study was conducted to fill this gap by reviewing all the relevant publications in 10 widely recognised journals in the field of technology-enhanced language learning. An analysis of 57 articles indicated five major types of technology for second and foreign language learning (i.e. technologies for mobile learning, multimedia learning and socialisation, speech-to-text and text-to-speech recognition, and digital-game-based learning). The results also showed four primary purposes and benefits of the state-of-the-art technologies: promoting practices, delivering instructional content, facilitating interactions, and restructuring teaching approaches. Moreover, these state-of-the-art technologies have been integrated into various aspects of language teaching and learning, the overall impact of which has been positive.

## KEYWORDS

Computer-assisted language learning; mobile-assisted language learning; mobile learning; multimedia learning; technology-enhanced language learning

## 1. Introduction

Technologies have been widely applied to the teaching and learning of various disciplines, including language education, and the quantity, quality, and diversity of the technologies that have been integrated into language learning are significant (Chen, Zou, & Xie, 2020; Slavuj, Meštrović, & Kovačić, 2017). At a time when the use of language learning apps is



**Figure 1.** Numbers of SSCI journal articles in the field of TELL.

steadily increasing, with smartphones and tablets now viewed as essential devices in language classrooms, and open-source language learning platforms and programmes mushrooming at an unprecedented pace, an increasing number of practices of technology-enhanced language learning (hereinafter, TELL) (Chen, Zou, Cheng, & Xie, 2020; Godwin-Jones, 2011) are appearing. Such practices continue to update our understanding of language education, with TELL, as a research field, becoming very active in recent years (Marijuan & Sanz, 2017). With the rapid development of digital devices and technologies, technologies are increasingly being used (in the classroom), leading to more diversified approaches to effective language education (Chen, Wang, Zou, Lin, & Xie, 2019).

For the present study, the researchers combed through Social Sciences Citation Index (SSCI) journals that are recognised in the field of TELL and ‘dedicated to technology integration in education’ (Hung, Yang, Hwang, Chu, & Wang, 2018, p. 93) for studies published on TELL. Figure 1 shows the number of SSCI journals in all years (from 1995 until the end of June 2019), retrieved from the Web of Science database using the search entry as ‘language learning’ and ‘technology’ or ‘computer’ or ‘mobile.’ The dashed line illustrates the overall ascending trend of journal articles on TELL published since 1995 as well as the remarkable increase of the publications in the most recent five years with almost half of the papers being published from 2016 to June 2019 (i.e. 466 out of 988 publications in total).

Along with the increasing interest and number of publications, several studies have been conducted reviewing the development of the field,

among which, seven appear most notable. The first systematic review was conducted by Golonka, Bowles, Frank, Richardson, and Freynik (2014) who reviewed 350 studies comparing the use of newer technologies with more traditional language learning methods or materials. They concluded that, generally, there was limited evidence for the efficacy of using technology in language learning. However, automatic speech recognition systems were viewed as effective in improving learners' pronunciation, and chat was conducive to an increased amount and complexity of learners' language production.

Slavuj et al. (2017) examined 42 TELL studies published in four journals (i.e. *CALICO Journal*, *Computer Assisted Language Learning*, *Journal of Educational Technology and Society*, and *ReCALL*) between 2005 and 2015, with a focus on adaptive and intelligent instructional systems and personalisation. They found that a variety of languages had been examined with English being the most frequently investigated. The results also showed that the intelligent tutoring system was the most common approach to adaptivity in language instruction. Specifically, there were three types of personalisation systems. The process-steering type adjusted the learning style and pace to learners' ability and preference; the content-adapting type delivered language knowledge based on learners' interests and proficiency levels; and the feedback-giving type providing in-time responses according to learners' learning behaviours.

Shadiev, Hwang, and Huang (2017) reviewed 37 journal articles published from 2007 to March 2016 on mobile language learning in authentic environments from four perspectives: publication trend, research focus, technology type, and research methodology. The results showed the dominance of English in mobile language learning research and revealed a rising academic interest in this topic. Smartphones and personal digital assistants were the most frequently used technologies and learners' language proficiencies and perceptions of mobile learning were the most popular topics. More importantly, Shadiev et al.'s findings indicated the potential of research on mobile technology-enhanced practical use of language knowledge in real-life situations.

Hwang and Fu (2018) conducted a systematic review of 90 articles on collaborative mobile learning in various disciplines. In a search of articles from the Web of Science, they selected the most relevant ones published between 2007 and 2016 and divided them into two periods to analyse research trends in mobile technology-supported collaborative learning. The review showed that most investigations of collaborative mobile language learning focused on learning English as a second language, while other languages received very limited academic coverage. The researchers also observed that most studies in this field focused on secondary and

university students as the target language learners, while adult, preschool, and kindergarten learners were largely ignored.

This three-step method of searching, selecting and dividing was also applied by Hung et al. (2018) in their review of 50 publications on game-based language learning that were published from 2007 to 2016. Surveying the designs and effectiveness of digital instructional games, the coding scheme of this research focused on research methods, types of digital games, participants and their target languages, and learning outcomes. The results, consistent with Slavuj et al. (2017) and Shadiev, Hwang, et al. (2017), showed that English was the most frequently investigated target language. The most popular game genres, which were normally integrated into learning tasks, were immersion games and multiplayer online role-playing games. It was also found that the participants of most studies were university students with mixed levels of language proficiency, and among various research issues, the impact of digital games on learners' psychological states was the most frequently examined. Most of the reviewed studies reported positive learning outcomes and affective states of students when learning with digital games.

Lee (2019) focused on the context-aware technology use in foreign language learning and reviewed 88 studies published between 2000 and 2018. The results showed that the academic attention on this topic kept fluctuating over the decade, and that most empirical studies in this field were either of one-session or short-term. Most researches employed quantitative methods and reported overall positive effects of the context-aware technology on learners' affective, linguistic, socio-cultural, and cognitive states, while learners' actual learning outcomes were seldom investigated. Lee therefore called for more pedagogy-informed long-term research on students' language learning achievements in context-aware technology-enhanced learning environments.

Shadiev and Yang (2020) conducted a comprehensive review of 398 TELL studies published between 2014 and 2019 from three aspects: the number of articles published by journals and by year, languages and skills, and technology used. The results showed that the number of publications was biggest in 2017 and in the *Computer Assisted Language Learning* journal. English and Chinese were the most frequently investigated target languages in TELL studies. Vocabulary knowledge and writing and speaking skills were the most frequently investigated aspects of language knowledge. The researchers also identify 23 types of technology that had been applied in language teaching and learning, among which digital games, corpus, and automated feedback were used most frequently.

These seven articles, similar to most review studies (e.g. Marijuan & Sanz, 2017), reviewed publications over a period of one decade.

However, studies that have been published after 2016 have yet to be reviewed in most review papers. Further, because of the recent rapid technical advances and innovations in modern technology, most reviews already appear dated in terms of the technologies reviewed within. For instance, the iPod, which is now close to obsolete as a learning tool, was regarded as an innovative device for language learning by Golonka et al. (2014). Thus, to provide researchers in the fields of CALL and TELL with an updated picture of the latest TELL developments and strategies, a review of the studies that were published after 2016 appears timely. Moreover, most previous TELL reviews have focused on one technology only: Hwang and Fu (2018) and Shadiev, Hwang, et al. (2017) on mobile technologies, Hung et al. (2018) on digital games, and Lee (2019) on context-aware technology, thus failing to provide a comprehensive understanding of a wide range of technologies for language learning in general. Additionally, many reviews on TELL focused mainly on the research methodologies of the previous studies (e.g. Hwang & Fu, 2018; Lee, 2019; Shadiev, Hwang, et al., 2017; Shadiev & Yang, 2020). They summarised the status of the field in terms of the publication years and journals, the target languages, the methods of experiments and data collection, thereby limiting the analyses of the types and purposes of the technology applied to enhance language learning and how such applications were conducted. As a result, there have been few, if any, in-depth reviews summarising the state-of-the-art technologies used to enhance second or foreign language learning, and analysing how such technologies have been used to promote effective learning.

To fill in these gaps, the present study reviews the publications in TELL in some widely recognised SSCI journals in the field over the period from 2016 to the first half of 2019, focusing on the types of the state-of-the-art technologies investigated and what roles they have played in language learning. Three research questions were proposed:

1. What state-of-the-art technologies have been used to enhance second or foreign language learning?
2. What were the purposes of integrating the state-of-the-art technologies into second or foreign language learning?
3. Were the state-of-the-art technologies effective in promoting second or foreign language learning?

## **2. Research method**

We identified and analysed relevant articles using a three-step process, article search and selection, and data analysis, following Hwang and Fu

(2018) and Zou, Huang, and Xie (2019). The search was conducted in Web of Science Core Collection, with '2016–2019' as the time-span, 'English' as the language, and 'article' as the required document type. Three keywords were used: 'technology,' 'language,' and 'learning,' with the AND operators between.

### **2.1. Journal selection**

For quality assurance, we selected articles that were published in 10 SSCI journals that are widely recognised in the field of TELL, as reliable review results are conditional upon reliable data sources. At an earlier stage of our review study, we had included book chapters, conference papers, and non-SSCI journal articles in the review list. However, after reading these papers, we found that the quality of many studies was unsatisfactory due to a lack of important details such as descriptions and explanations of what technology was used, as well as how and why such technology was applied in the research. Moreover, some studies did not implement a purposeful use of technologies for language learning, but rather investigated technology for its own sake. For instance, a conference paper reported that it used hyperlinks to facilitate reading but did not include details concerning how hyperlinks were integrated in reading materials, what content the hyperlinks provided, and why such content may assist reading. Another non-SSCI journal article claimed that it investigated the effects of computer-assisted and mobile-assisted reading but did not collect or analyse any data concerning the participants' reading performance. These studies are anonymised for respect for the researchers. Since the aim of the present study is to review the types, purposes, and effectiveness of the technologies used in various TELL studies, prerequisites for the selected research articles include not only purposeful use of technologies for language learning but also detailed information concerning the types and effectiveness of these technologies; otherwise, the research questions can hardly be well addressed. Therefore, we decided to limit our selections to publications of high quality.

We selected 10 journals according to the SCImago Journal Rank (SJR) which ranks journals based on scientific indicators developed from the Scopus<sup>®</sup> database. First, we identified four journals that publish research on educational technology out of the top 35 journals in Quartile 1 in *Language and Linguistics*, according to the SJR 2018. They are *Computer Assisted Language Learning*, *Language Learning & Technology*, *System*, and *ReCALL*. Second, we identified five journals that publish research on language education out of the top 12 journals in Quartile 1 in *E-learning*,

namely, *British Journal of Educational Technology*, *Computers & Education*, *Educational Technology & Society*, *Interactive Learning Environments*, and *Journal of Computer Assisted Learning*. In this way, we selected nine journals. Then, we compared our journal list to the lists of other review studies, finding that Hung et al. (2018) and Hwang and Fu (2018) selected the same nine journals, but they also included a tenth journal, *Educational Technology Research & Development*. Considering that this is an influential journal in the field of educational technology and language education, we decided to include it in our list as well. Thus, we finalised 10 journals for the selection of articles for review.

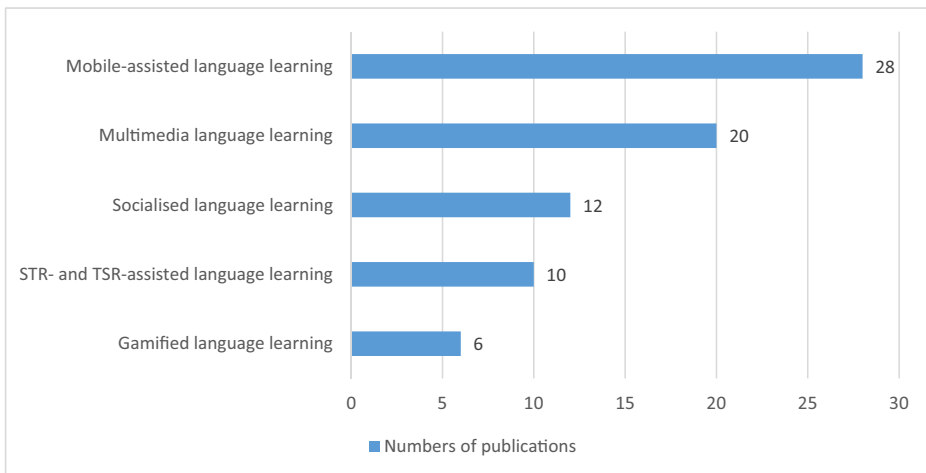
## 2.2. Article selection

The present search was conducted at the end of June 2019 generating 220 articles, which were further examined based on the following inclusion criteria. First, the publications had to be original research articles that investigated technology-enhanced second or foreign language learning. Second, the research had to be empirical; and third, quantitative data on the effectiveness of technologies in promoting language learning had to be analysed. Two researchers independently reviewed all 220 articles based on the criteria. Differences were resolved between the two raters via discussion, and the inter-rater reliability was higher than .94. One hundred and sixty-three articles were excluded at this stage, examples of which included Li and Chu (2018) for centring on first language learning, Slavuj et al. (2017) for being a pure review paper and Balaman (2018) for focusing on learners' psychological states rather than language learning outcomes. The 57 articles that were finalised for the review are listed in [Appendix A](#).

## 2.3. Data analysis

To address the research questions, the selected articles were analysed from three aspects: the types, purposes, and effectiveness of the state-of-the-art technologies. The two researchers examined five articles together first in order to generate initial codes. In case of any differences, the two researchers re-read the papers and discussed the coding until agreement was achieved. Differences were resolved via discussion for four of them, and an informed third rater was consulted for one paper where the initial two raters could not reach agreement. The types, purposes, and effectiveness of the technologies were then coded literally as reported by the research articles. After finalising the coding scheme which was used to systematically review the 57 selected articles, the two raters





**Figure 2.** Five types of state-of-the-art technologies for language learning.

independently coded the remaining articles, after which they compared their coding results. At the end of the independent coding, a satisfactory agreement was achieved (Pearson's  $r = 0.90$ ) with differences resolved via discussion.

### 3. Results

Our coding exercise generated five main types and four major purposes of the state-of-the-art technologies for second or foreign learning, and these are aligned with the sub-sections of this section; generally, positive findings concerning the effectiveness of these were noted.

#### 3.1. Types of the state-of-the-art technologies for L2 learning

As illustrated in [Figure 2](#), 28 out of 57 publications investigated mobile-assisted language learning; 20 investigated multimedia language learning; 12 investigated socialised language learning; 10 investigated speech-to-text recognition (STR) and text-to-speech recognition (TSR) assisted language learning; six investigated gamified language learning. The total number was bigger than the sum, 57 because some studies investigated more than one type of the state-of-the-art technologies. For example, Yukselturk, Altıok, and Başer (2018) examined both Kinect camera and game-based learning technologies. Tseng, Lin, and Chen (2018) developed an immersive online language course involving mobile apps (i.e. *Padlet* and *Flipgrid*), online platforms (i.e. *Zaption* and *Quizlet*), and multimedia technologies (i.e. videos and audio clips).

### 3.1.1. Technologies for mobile learning

Mobile learning refers to learning with mobile devices, such as smartphones, tablets, digital pens, and wearable devices like smartwatches, AR and VR glasses (Huang, Yang, Chiang, & Su, 2016). In our review, the integration of mobile technologies into language learning was the most frequently investigated in TELL. The advantages of mobile technologies included their high accessibility, supporting various learning styles (self-paced, collaborative, team-oriented, etc.) [and] enriching the static content with narratives, game-playing, hands-on activities, etc.’ (Subramanya, 2014, p. 39).

Our review found that the use of mobile applications was the principal approach to mobile learning. Researchers normally chose commercial apps or self-developed ones and asked participants to download them for language learning. For example, Chen, Carger, and Smith (2017) developed an English narrative writing course with Penultimate, an iPad-based commercial app that supports digital handwriting to facilitate the development of students’ writing skills. Their results showed that this app was effective in improving students’ narrative writing abilities and learning attitude. Lan and Lin (2016) developed an application, MOBILE SEamless (MOSE), to create real-world contexts and facilitate pragmatic speaking. The results indicated that MOSE users had significantly fewer errors when executing language tasks and far more interactions with their peers than traditional learners. Some studies examined the use of mobile devices’ basic functions. For example, Liu (2016) examined the use of the Instant Message function (IM) of mobile phones in learner-constructed concept-mapping for word learning.

In addition to the above examples, mobile technology has been widely applied in blended use with other technologies. Chang, Warden, Liang, and Chou (2018), for example, blended ubiquitous learning outdoors with computer-assisted learning indoors by integrating GPS into mobile devices and found the proposed system effective in enhancing listening comprehension and reducing learners’ cognitive load. Another example was a study by Liu, Huang, and Xu (2018), who integrated emotion recognition technology into mobile language learning. They developed a mobile-based application, *Story and Painting House*, which supported a free-space digital storytelling approach to enhance learners’ accuracy and creativity of speaking. In the storytelling activity, *FaceReader*, an emotion recognition software, was used to capture students’ facial expressions, analyse their anxiety levels, and keep teachers and researchers updated with information about their students’ emotions. Positive results were reported in this research.

### 3.1.2. Technologies for multimedia learning

Multimedia-assisted language learning refers to presenting language knowledge (e.g. grammar knowledge and word concept) or conducting

learning activities (e.g. exercises and instructional games) using multimedia, such as videos, audios, and images (Aldera & Mohsen, 2013). In one study, appropriate use of multimedia was conducive to learning by deepening knowledge processing, facilitating retention, and promoting motivation (Hwang, Shadiev, Hsu, Huang, Hsu, & Lin, 2016). Okuno and Hardison (2016) compared the effectiveness of auditory-visual and auditory-only systems in L2 pronunciation training and found that the combination of audio and video was the optimal media for vowel training. Both Chen, Tan, and Lo (2016) and Liakin, Cardoso, and Liakina (2017) reported positive learning outcomes when using Moodle, an interactive multimedia learning software.

Some studies have also investigated the integrated use of technologies for multimedia learning and other types of technology. For instance, Liu, Chen, and Hwang (2018) developed a collaborative e-learning system using video-based materials and tools for context-aware ubiquitous learning. The participants learned in a fitness centre, and their listening and speaking proficiency level improved significantly. Park and Warschauer (2016) developed a laptop-based textbook that could visualise texts and syntactic structures using multimedia technology and visual-syntactic text formatting technology. The results showed that this digital textbook was effective in enhancing students' grammar knowledge development, especially for low-proficiency learners who had weaker syntactic processing skills.

### 3.1.3. *Technologies for socialised learning*

Socialised learning refers to learning with socialisation tools such as online platforms or communities and social networks (Çetinkaya & Sütçü, 2018). Online platforms can support learning activities (e.g. discussion forums for information exchange) and provide instructional content (e.g. *Edpuzzle* and *Quizlet* for language input), and social networks create online communities for communication (e.g. *WhatsApp*, *LINE*, and *Facebook*) (Çetinkaya & Sütçü, 2018). Both online platforms and social networks have become an integral part of modern life and were widely applied in many of the reviewed papers to facilitate blended and flipped learning (Liu, Tai, & Liu, 2018). Lin and Hwang (2018) investigated the use of *Facebook* in an online language classroom and found it very effective in improving learners' oral skills and increasing their peer-to-peer interactive behaviours both inside and outside the classroom. Yang (2016) developed an online learning platform, *ESP Café*, in which the discussion forum was one of the primary components. Language students were encouraged to discuss the knowledge points with their peers and tutors.

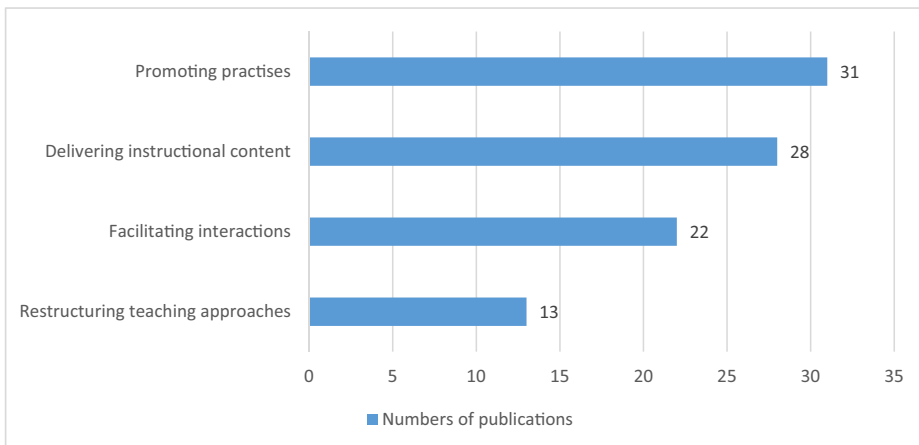
### 3.1.4. Technologies for speech-to-text recognition and text-to-speech recognition

STR and TSR refer to the technologies that automatically generate the textual or audio counterparts of learners' speech (or writing). The literature indicates that they are effective in deepening learners' knowledge processing and memorisation by strengthening the connections between audio and textual forms of language knowledge (Mayer & Moreno, 2003; Shadiev, Hwang, & Liu, 2018). For example, Shadiev, Hwang, and Liu (2018) integrated both STR and TSR into a smart-watch-based learning system. During speaking training, STR was used to recognise learners' errors and generate text for their correction; and in writing exercises, TSR was applied to help learners avoid grammar and spelling mistakes by speaking out the suggested versions. The results showed that the participants who experienced the STR- and TSR-enhanced training had very effective learning outcomes and positive perceptions of the learning system. Bodnar, Cucchiarini, Penning de Vries, Strik, and van Hout (2017) developed a corrective feedback (CF) system that assisted oral grammar learning by automatically recognising and pointing out errors in learners' speech. Although the results showed no significant effect of the CF system, the great potential of automatic speech recognition technology in promoting oral grammar practice was observed and reported.

Nguyen, Hwang, Pham and Ma (2018) integrated STR technology with cloud translation technology and developed a mobile application called *ezTranslate* to automatically recognise and translate learners' speech, utter the translation, and permit learners to repeat after the utterance. The results showed that learners' overall speaking proficiency was enhanced after using *ezTranslate*.

### 3.1.5. Technologies for digital-game-based learning

Game-based learning refers to the integration of game elements (e.g. rules and goals, challenge, mystery/storyline, rewards, etc.) in delivering instructional content and conducting learning activities (Yukselturk et al., 2018). Language courses using game environments promoted learners' engagement, triggered their interactions, and enhanced their knowledge comprehension (Hung et al., 2018). Notable among the reviewed studies in this area, Rachels and Rockinson-Szapkiw (2018) designed a game-based Spanish course to enhance primary students' general language skills. The games were based on Duolingo<sup>®</sup>, a free application which can personalise learning experiences by providing learners with appropriate scaffolding and adapting the instructions according to their language proficiency levels. Similarly, Hung (2017) gamified in-class learning activities using *Kahoot!*, another free application that can be



**Figure 3.** Four major purposes of the state-of-the-art technologies.

used to create instructional games. With *Kahoot!*, the class was turned into ‘a game show in which the teacher becomes the host and the students the contestants,’ and learner motivation, engagement, and interactions among teachers and peers were increased (p. 986).

Some studies on game-based language learning integrated other technologies such as Kinect technology and augmented reality (AR). For example, Yukselturk et al. (2018) developed three games that immersed learners in game environments where they took active roles in learning while interacting with others. Kinect technology was applied to enhance learners’ game experiences. A motion sensing input device consisting of an RGB camera, three-dimensional depth sensors, a multi-array microphone, and built-in processing cores were used to give synchronous commands based on learners’ responses. It was found that the game players achieved significant development in speaking, listening, reading and writing. Hsu (2017) also developed a task-based, AR-enhanced game system to promote situated vocabulary learning. Enhanced by AR, the game created a three-dimensional space where learners received and applied language knowledge in virtual and real worlds. The results showed that the students who learned with AR achieved better learning outcomes than those who did not.

### **3.2. Purposes and benefits of the state-of-the-art technologies**

As shown in Figure 3, 31 of the articles reported on the effectiveness of the state-of-the-art technologies in promoting practices, 28 on delivering instructional content, 22 on facilitating interactions, and 13 on restructuring teaching approaches. The total number was bigger than the sum, 57 because most technologies were multi-purposeful. Tables in Appendix B

illustrate the effects of different types of the state-of-the-art technologies on various aspects of language learning outcomes in language development and psychological states. [Appendix C](#) shows detailed information about the types and functions of technologies, and research results of the 57 papers.

### 3.2.1. *Promoting practices*

Technologies have been effective in promoting language practices and learning behaviours that involved unsolicited uses and reuses of target languages in context (Rampton, 2002). [Table B1](#) in [Appendix B](#) illustrates the types of technology used among the 57 articles to enhance the development of different language-knowledge aspects by promoting practices and the effects of these technologies.

Mobile technology was the most frequently investigated innovation for promoting practices. With unprecedented affordability and accessibility, mobile technology minimises hardware, as well as the conventional temporal and spatial restrictions that sometimes impede instructional content and learning activities, and therefore maximises students' exposure to the learning environment (Darling-Aduana & Heinrich, 2018). For example, Kurt and Bensen (2017) proposed a vocabulary knowledge practising system based on *Vine*, a mobile application that allows learners to record and share 6-s-long video clips. Learners enhanced their vocabulary knowledge by speaking and spelling via their mobile devices and reviewed their knowledge by looping videos. The results of the pre- and post-tests and interviews showed that the *Vine*-based practice was effective in promoting participants' motivation to learn as well as improving their vocabulary and speaking. Based on the results, the researchers predicted that there would soon be a broader adoption of mobile and multimedia technologies into vocabulary classrooms. Liu (2016) attempted to facilitate learners' practising of vocabulary knowledge through the Instant Message function (IM) of smartphones. In this study, the experimental participants practised their vocabulary knowledge by creating multimedia content via IMs, while the control participants learned words in the traditional way without any electronic support. The pre- and post-test scores of the two groups showed that the former group had significantly better learning outcomes, indicating that the IM-based concept-mapping was conducive to vocabulary acquisition and retention. However, the delayed post-test revealed that mobile-assisted learning may be effective in the long term only when effective learning strategies are employed.

For the purposes of encouraging practices, gamified learning was also frequently investigated and applied to trigger learners' motivation and

provide interesting contexts of knowledge use. For instance, Castañeda and Cho (2016) established practising contexts on mobile-based gamified language learning, by requiring their participants to assess and practice their accuracy in conjugating Spanish verbs through games. The results of pre- and post-tests showed that the game-based practices had positive effects on the learners' Spanish grammar knowledge. The participant interviews and questionnaires also indicated their increased confidence in the proposed language learning activity. The researchers claimed that the mobile-based gamified learning helped the students acquire complicated grammar knowledge by providing a competitive, but enjoyable environment in which they could 'intuitively apply' the target knowledge with immediate feedback through discussions with their peers (p. 1202). Similarly, Hsu (2017) developed a digital game integrating Augmented Reality technology through which learners could have a vivid application of word knowledge within simulated contexts. Compared to those who had practiced using traditional methods, the language learners using the AR-enhanced game were found to have higher levels of learning anxiety and mental effort but performed significantly better in the post-tests of vocabulary knowledge. The researcher postulated that the proper amount of learning anxiety and mental effort required in playing the AR-enhanced instructional game may prompt their learning motivation and thereby have resulted in the positive learning outcomes. However, not all attempts to encourage practices through gamified learning ended with success. Rachels and Rockinson-Szapkiw (2018) required one group of participants to practice their vocabulary and grammar knowledge on a Duolingo-based instructional game and compared their learning outcomes to another group of participants who had practised in a traditional way. The comparison revealed that the proposed gamified learning increased motivation, but had little effect on enhancing vocabulary and grammar learning. The reason for the limited effect was postulated to be the similarity between the Duolingo-based learning and the traditional face-to-face instruction in providing scaffolding to the learners. Because of the similarity, the researchers speculated that the Duolingo-based learning could be used in schools to replace teachers.

Some other state-of-the-art technologies were also investigated for enhancing language learning through promoting certain practices. Tan, Chen, and Lee (2019) developed a multimedia computing platform, *Livescribe*, which allowed learners to practice listening exercises by tapping an interactive board with digital pens. The researchers analysed several aspects including: the platform operating logs, a questionnaire about the participants' motivation, a metacognitive awareness questionnaire, and two sets of pre- and post-listening comprehension tests. The results



showed that the proposed practising method significantly improved learners' listening comprehension and motivation. Based on an analysis of the success of *Liverscribe*, Tan et al. (2019) stated that the learners could 'show off' their practising scores, share knowledge and have positive competition with their peers, eventually achieving better learning outcomes if there was a public practising environment. The researchers therefore suggested good potential existed for educators and researchers to further explore in this direction. Ho, Hsieh, Sun, and Chen (2017) developed a Ubiquitous Learning Instruction System with Augment Reality features (UL-IAR), attempting to facilitate learners' English listening and speaking practices with 'the concurrence of digital information and real environments' (p. 177). In order to cater to the students' different cognitive styles of learning and practising, Ho et al. (2017) designed the AR-featured system based on the theory of field dependence and field independence. With this system, the students who preferred having guidance during their learning processes could learn with the structured learning materials and practice with 'the stimulus of external information and interactivity,' and those who preferred independent thinking were provided with internal references and encouraged to have 'analytical and active learning' (p. 179). By comparing the performances of the control and experimental groups, the researchers found positive effects on the learners' listening and speaking, and implied that the AR-featured ubiquitous learning system would lead to even better learning outcomes if it were integrated with effective learning strategies.

### 3.2.2. Delivering instructional content

Content in language learning, which refers to knowledge about language, plays an important role in educational programmes; and the approach to content delivery is to some extent a determinant of how learners process the content (Chen, Wang, & Lin, 2015). Table B2 in Appendix B illustrates the types of technology used to enhance the development of various language knowledge aspects through delivering instructional content, in addition to the effects of these technologies.

Multimedia was the most frequently used type of technology for content delivery. It was embedded in storytelling in several studies, with Mayer's cognitive theory as the theoretical foundation (e.g. Hwang et al., 2016; Shadiev, Huang, & Hwang, 2017). According to Mayer and Moreno (2003), when content is delivered via multimedia, learners have both 'auditory/verbal' and 'visual/pictorial' channels activated and can achieve better understanding and learning (p. 44). Liu, Huang, et al. (2018), Hwang et al. (2016) and Liu, Wang, and Tai (2016), for instance, applied the digital storytelling system to present instructional passages,



where multimedia was used either as presentation media or hypermedia. In these investigations, the participants were divided into different groups and learn languages based on different types of instructional media. The performances in pre- and post-tests by the participants of different groups were then collected and analysed. Most participants in these investigations were able to process the target information in the two memorising channels simultaneously with the help of audio and video illustrations and clips and achieved positive outcomes in speaking, listening and vocabulary (Hwang et al., 2016; Liu, Huang, et al., 2018; Liu et al., 2016). Hwang et al. (2016) also suggested that the animations embedded in the storytelling system would be particularly effective in helping primary school students to attain higher levels of learning motivation and longer retention of the vocabulary knowledge. Connor et al. (2019) developed a tablet-based multimedia e-book with comprehension monitoring technology that could monitor learners' progress and comprehension of the instructional contents and provide personalised learning experiences. The results of experiments among 603 primary students showed that the proposed e-book largely assisted students' development of word knowledge and learning strategies. The researchers suggested that the proposed e-book could help young learners not only in vocabulary acquisition, but also in the improvement of judgements in reading and learning. However, not all the results indicated that the use of multimedia in language instruction had positive effects. For example, Lin (2017) required one group of participants to learn the targeted language knowledge on tablet-based multimedia textbooks and the other to learn in a traditional way. No significant differences were found between the two groups of learners in the post-training tests of reading comprehension. Based on the questionnaire results, Lin postulated that the negligible effect might have resulted from the learners' long-term learning habits of paper-based note-taking. The participants were used to reading printed textbooks and relied heavily on hand-written notes they took in class so they had difficulty adjusting to the new reading medium. In addition to embedding multimedia in storytelling, some studies have integrated multimedia into language learning systems. For example, Wang, Lin, Hwang, and Liu (2019) applied multimedia instruction in a contextualised learning environment by utilising sensing technology with the mobile app, *GBELA*. Learners received different sets of audio-visual language materials and learning tasks depending on their contexts as sensed by the app. The results showed the effectiveness of this system in enhancing learners' L2 listening and reading skills as well as their self-efficacy. Based on the success of *GBELA*, the researchers suggested that the learning materials and instructional activities in multimedia apps should be developed according to the different goals of target learners.

TSR and STR were also frequently applied to facilitate content delivery. Shadiev, Huang, et al. (2017) integrated lexical and audio forms of language knowledge by employing an STR system upon language learning texts. The results of pre- and post-tests revealed that participants who learnt with the proposed system outperformed those who did not and showed more positive learning attitudes and higher levels of learning attention indicating multimedia rendered the content to be ‘more comprehensible and easier to process’ (p. 1257). Based on the positive results, the researchers suggested that educators and researchers be aware of the potentially wide applications of the STR system to the language learning classroom, especially for male students and students of low language proficiency who were found to have progressed most in the post-tests. In another experimental study (Huang, Shadiev, & Hwang, 2016), participants in an experimental group were provided with a computer-based STR system in L2 lectures, while a control group was not. The results showed the experimental group had a significantly better improvement in L2 speaking and had a lower cognitive load. However, the STR system seemed more useful for the learners of low language proficiency than those of high proficiency. Therefore, the researchers called for educators and researchers to pursue further improvement of the STR system in order to help learners of differing levels receive appropriate multimedia lectures in accordance with their language levels. As for TSR modelling, Liakin et al. (2017) combined a TSR system with Moodle for teaching students French pronunciation. This long-term study indicated that although little difference was found in the immediate learning performances of the experimental and control groups, only the former one achieved progress over time. They suggested that the proposed TSR system could increase the ‘quantity and quality’ of learners’ input and can be a very productive tool for conducting repetitive pronunciation activities out of class (p. 336).

### ***3.2.3. Facilitating interactions***

Language learning is a social process in which learners acquire and apply the target language knowledge in their interactive communications with peers and tutors (Mitnik, Recabarren, Nussbaum, & Soto, 2009). Through interactions, students reflect on what they have learnt, connect new knowledge elements within an overall system, share their achievements and experiences ‘of various levels of expertise’ and eventually obtain new models and perspectives for future learning (Shadiev, Hwang, Huang, & Liu, 2018, p. 44). Table B3 in Appendix B illustrates the types of technology used to enhance the development of different language

knowledge aspects through facilitating interactions, in addition to the effects of these technologies.

The review found that technologies were mainly used to facilitate two types of interactions: peer-to-peer and teacher–student communications and peer-to-peer collaborations. Despite some overlap, the former emphasised idea-exchanging, problem-solving and experience-sharing in and out of class, while the latter focused on students’ face-to-face or web-based teamwork upon instructional activities or learning tasks (Mitnik et al., 2009).

In an attempt to facilitate the in-class peer-to-peer and teacher–student communications, Hung (2017) proposed a game-like clicker application with which students could raise questions and express opinions synchronously using mobile devices. Based on the data collected through post-tests, participant interviews and questionnaires, the results showed that the proposed application helped learners improve their learning performance and acquire positive learning attitudes. The researcher argued that the positive outcomes may have resulted from the gamification in the technology-enhanced learning activity and the scaffolding and interaction opportunities throughout the learning process, while advocating the broader use of the clicker in the language classroom. Roussel and Galan (2018) also used clickers to expedite learners’ self-reporting and opinion-voicing in vocabulary classrooms. The results showed that learners who used clickers to learn had a better understanding of the lecture content, higher efficiency of language learning and a lower cognitive load, especially when the learners were of low language proficiency. To maximise the effectiveness of the clickers, the researchers suggested that course content should be closely connected with the target knowledge, and they noted ‘a high level of interactivity between the elements that compose it’ (p. 54). In another study aimed at facilitating in-class communications, Tsai (2019) created interactive courseware which allowed learners’ construction and integration of ‘verbal and visual cognitive representations’ when learning to write in their L2 (p. 46). This courseware was effective in improving learners’ writing skills and learning attitudes, as demonstrated by the participants’ pre- and post-writing performance and questionnaire. Tsai suggested the proposed learning courseware would be particularly useful for shy students by helping them receive face-to-face instructions from the teachers without asking questions in public. Similarly, Song, Oh, and Glazewski (2017) developed a customised personal response system through which the instructor raised questions online for students to find the answers through in-class discussion. Students could also easily generate and post their own questions through the system for their peers to solve. The researchers found that the

proposed system could facilitate learners' development of grammar knowledge and communication skills, increase their engagement with the learning activities and largely enhance in-class interactions. Based on the positive results, the researchers recommended a wider application of this system for supporting student-generated questioning. However, they also stressed that this system may be more suitable for high proficiency students than the low ones because it was 'related to a higher extent of instructional goals and topic' and could be challenging for the low-proficiency students (p. 1444).

As for out-of-class communication, social networks were effective in prompting peer-to-peer information sharing and opinion exchanging. Çetinkaya and Sütçü (2018) investigated the possible advantages and problems associated with using WhatsApp and Facebook as interaction platforms for students' vocabulary learning and found that WhatsApp facilitated effective learning, while Facebook did not. Based on their investigation, the researchers highly recommended the use of social networks in vocabulary courses and suggested the course instructors become involved in the socialised learning process. They argued that future studies in this field could focus on students' perceptions and opinions of the interactions for language learning. Yeh and Lai (2019) used Skype as a tutor-student communication tool through which participants were required to have discussions with their peers and receive online-tutoring for nine weeks. Their analyses of the pre- and post-tests of the participants' oral performances indicated that Skype-assisted interactions were conducive to learners' L2 speaking development, especially among those of low language proficiency. The researchers suggested that future studies in this field could generalise these results using a larger sample over a longer duration, and focus on tutors' feedback in the Skype-based interaction system, especially on the 'relationship between students' outputs and tutors' feedback and instructional strategies' (p. 190).

Collaborative tasks are generally in the format of 'an activity where there is a shared and negotiated decision-making process and shared responsibility for the production of a single text' (Storch, 2013, p. 3). Collaborative learning plays a vital role in language learning, and various technologies, especially internet technology, have been applied to facilitate it. Bikowski and Vithanage (2016) required experimental participants to complete collaborative L2 writing tasks using Google Docs and found that they had statistically significant writing gains as well as very positive attitudes towards technology-enhanced collaborative learning. This study therefore revealed the great potential of in-class web-based collaborative writing tasks and suggested a three-stage teaching cycle for conducting this learning method: (1) preparation, (2) collaborative writing, and (3)

reflection. Talaván, Ibáñez, and Bárcena (2017) designed a collaborative reverse subtitling task based on *Aegisub*, a sub-title-making software, and reported significant effectiveness of the technology in promoting learners' writing, especially for coherence, cohesion and idea structuring. This effectiveness may have resulted from the design of video interfaces that guided the learners' attention to the 'discursive aspects' of the learning process and the 'communicative and functional' nature of the subtitling activity (p. 55). They recommended future studies in this field could investigate the learners' preference for studying collaboratively as well as their giving of feedback during the learning process.

### 3.2.4. Restructuring teaching approaches

The development of technology and change of lifestyles can uncover the transition of instructional structures from the fixed to the flexible and the teacher-dominant to the learner-centred, eventually, leading to more efficient approaches (Bergmann & Sams, 2012). By supporting contextualisation and personalisation of instruction and access to information repertoires, technologies enable hypothetical pedagogical approaches to be realised and drive their further specification and development (Huang et al., 2016). The present review reveals various attempts to explore the use of different technologies to improve instructional structures and pedagogical approaches in the field of TELL. Table B4 in Appendix B illustrates the types of technology used to enhance the development of different language knowledge aspects by restructuring teaching approaches, including the effects of these technologies.

The flipped classroom was the most frequently investigated structure enhanced by technologies. This approach to instruction refers to the reversion or flipping of conventional pedagogical processes (Bergmann & Sams, 2012). In this structure, the content delivery, which is traditionally conducted through teachers' in-class lectures is moved out of the classroom into students' own negotiated time and place; and as a replacement, interactive learning tasks and activities are moved into the class (Lin & Hwang, 2018). This teaching approach can enhance teacher-student communication and mutual understanding, increase learners' motivation and engagement in learning activities, and improve learning efficiency (Hsieh, Wu, & Marek, 2017). With the fast development of multimedia, mobile and Web technologies, flipped learning has been widely applied in the language learning area in recent years, and positive learning outcomes have been commonly reported. Lin, Hwang, Fu, and Chen (2018), for instance, proposed a flipped learning approach based on a computer-mediated contextualisation system and instructional games, and their log file data indicated that the experimental group had

fewer errors in business writing tests and higher motivation than those who learned in a conventional classroom. The researchers suggested that future studies in this direction could investigate the effectiveness of the proposed learning approach in language learning courses other than business English, while exploring students' higher-order thinking, self-efficacy, learning anxiety and interactions with the tutors in the computer-mediated flipped classroom. Hsieh et al. (2017) used *LINE*, a mobile communication application, to enhance a flipped oral training course where the participants were able to practice target knowledge beyond the classroom through written and verbal discussions. The results indicated that the flipped learning approach was conducive to L2 idiom learning and that *LINE* could be used to support this approach. To effectively implement this approach, the researchers suggested teachers explain the instructional design and the formative benefits of flipped learning at the beginning and monitor the students' progress throughout the learning process in case some students do not understand the meaning of the approach or cannot regulate themselves to keep up with the learning pace. Similarly, Haghighi, Jafarigohar, Khoshshima, and Vahdany (2019) based their design of flipped learning on the mobile communication app, *Telegram*, and reported considerable improvement of the participants' L2 pragmatic competence and learning satisfaction. The researchers argued that this significant improvement might have resulted from the compatibility of *Telegram* with the structure of the flipped learning and the increased opportunities of knowledge application and interactions that the flipped classroom provided. Shyr and Chen (2018) proposed a computer-based system, *Flip2Learn*, to facilitate the out-of-class learning part of their course and encouraged teacher-student communication by a self-regulation and guidance system. The results indicated that the proposed flipped system was effective in enhancing language learning performances and motivation. Theoretically, this study re-confirmed the effectiveness of the flipped learning approach in language learning and revealed the positive influence of the self-regulation and guidance system on deepening the learners' processing. Practically, the researchers suggested language teachers 'train their students to regulate their learning with hypermedia' before conducting the flipped approach pedagogy (p. 60). Wang, An, and Wright (2018) developed a MOOC-based flipped classroom that saved class time from teacher-led instruction and allowed the students to orally discuss and communicate with their peers in an EFL class. Compared to the learners who learnt speaking skills in traditional classrooms, the learners in their flipped classroom learned the instruction much more quickly, demonstrated more positive learning attitudes and had significantly more progress in

speaking proficiency, especially in speech fluency. Although the merits of the proposed learning method seemed to decrease over the long term, especially in speaking accuracy and complexity, the researchers concluded that, overall, the MOOC-based flipped learning was a promising approach to enhancing oral language proficiency and called for further explorations in this field to fix the two challenges.

Some studies attempted to restructure teaching approaches from the perspective of instructional foci. The traditional language classroom tends to have meaning-focused instruction where language concepts are taught in isolation occupying the majority of learners' and tutors' attention while language contexts and forms are 'by and large sidelined or trivialized' (Marefat & Hassanzadeh, 2016, p. 107). Marefat and Hassanzadeh designed a set of computer-mediated form-focused language lessons that drew learners' attention to the forms and structures of the language within the context of communicative interactions by giving metalinguistic information, highlighting the form and providing corrective feedback. The results showed that the form-focused language teaching method led to better learning efficiency by facilitating learners' vocabulary acquisition in corresponding contexts. Despite the positive results, the researchers noted that computer-mediated form-focused language learning was 'an intricate practice' and deserved more investigation before wider implementation (p. 120). They also suggested that language teachers should pay more attention to decontextualised learning practices and the long-term merits of this learning method. Similarly, Huang and Chuang (2016) emphasised the significance of authentic content and contextualisation in language learning while adopting content-based instruction theory and developing an animation-based reading course. Their results showed that the participants in their experimental group outperformed those in a control group from the perspectives of reading speed, technology acceptance and attention on the target knowledge. They claimed that with the proposed teaching approach, the teachers may have better control over the learning conditions in a large class which can help students, especially those with low proficiency, achieve better learning outcomes.

#### **4. Discussion and conclusion**

The present review of research on TELL from 2016 to June 2019 indicates there has been a broad application of various state-of-the-art technologies in language learning. In answering the three research questions, the review reveals that among different technologies, mobile learning, multimedia learning, socialised learning, STR and TRS recognition, and



digital-game-based learning were the most frequently investigated in the literature. These technologies were mainly used to promote language learning practices, deliver instructional content, facilitate interactions, and restructure teaching approaches. Also, in general, positive results concerning the learning outcomes and motivation of the students who learned with technological support were reported.

#### **4.1. Pedagogical implications**

For the purposes of promoting practices, educators and researchers were advised to apply mobile technologies, including mobile-based multimedia applications, such as *Story and Painting House*, *Penultimate*, and *Vine*, and mobile handwriting tools, such as digital pens (Chen, Carger, et al., 2017; Kurt & Bensen, 2017; Liu, Chen, et al., 2018; Tan et al., 2019). Digital game elements were frequently integrated into mobile-assisted language learning and were found to be effective in improving learning experiences prompting learner motivation.

For educators and researchers who want to enhance language learning in the direction of instruction delivery, multimedia was a recommended choice. The animations and audio and video clips embedded in storytelling systems were viewed as facilitative for learners' development of speaking, vocabulary, and comprehensive language skills as well as for improving their learning motivation (Hwang et al., 2016; Shadiey, Huang, et al., 2017).

The review also revealed that educators and researchers can apply technologies to facilitate interactions from two dimensions. For peer-to-peer and teacher-student communications, clickers and social networks were found effective in bridging the in-class and out-of-class activities seamlessly, expediting the question-and-answer interplay between students and instructors and encouraging students to apply their newly acquired language knowledge (Çetinkaya & Sütçü, 2018; Hung, 2017; Roussel & Galan, 2018; Yeh & Lai, 2019). For increasing peer-to-peer collaborations, *Google Doc* and *Aegisub* were viewed as tools that encourage learners to have group discussions and conduct writing and translation exercises in teams (Bikowski & Vithanage, 2016; Talaván et al., 2017). The review showed that most learners could improve their ability in speaking, listening, writing, vocabulary, and learning motivation through technology-enhanced interactions with their peers and teachers.

In restructuring teaching approaches, some articles recommended the flipped classroom model based on the combined use of social networks, multimedia, Internet, and mobile technologies. *LINE*, *Telegram*, and *Flip2Learn* were investigated and found to enhance learning motivation



and the learning of speaking, vocabulary, and comprehensive language skills by increasing learners' interactive learning activities with their classmates and teachers (e.g. Hsieh et al., 2017; Lin et al., 2018; Shyr & Chen, 2018).

#### **4.2. Future directions for TELL**

Our review has identified similar findings with previous review studies, one of which is the types of commonly applied and investigated language learning technologies. Consistent with Golonka et al. (2014), we found that mobile devices and social networks are widely applied in TELL. We also identified four most frequently used technologies, namely the digital game, social networks, STR recognition and multimedia technologies, which were consistent with the findings of Shadieff and Yang (2020). However, different from their review which summarised the content of the reviewed studies from the perspectives of the article counts by journals and by year, the target languages and skills, the types of technology, and the trend, our review involved more in-depth analyses of the types, purposes, and effectiveness of the state-of-the-art technologies. The findings that have been identified in both our review and previous reviews indicate the effectiveness and technology maturity of mobile learning, socialised learning, digital-game-based learning, multimedia learning, and STR recognition for TELL. Future educators and researchers are therefore encouraged to make full use of these state-of-the-art technologies to enhance language learning.

In addition to the above similar findings, our review also found new developments and tendencies in TELL. Some types of technology that were popular several years ago have rarely been discussed or used in recent years, while others have experienced great developments recently. For example, computer-mediated chats and iPods were frequently used to increase learners' interactions and facilitate their mobile learning when Golonka et al. (2014) conducted their review. Yet neither was investigated in the 57 articles that we reviewed. As replacements, social networks, smartphones, and tablets have been extensively applied in recent three years (e.g. Chen, Carger, et al., 2017; Lin & Hwang, 2018). Compared with the chats, social networks enable learners to have group discussions, post and repost learning information, and make comments in addition to one-to-one talks (Satar & Özdener, 2008; Sykes, 2013). Learners can also interact using pictures and videos, apart from texts and audios. Compared with iPods, smartphones and tablets have more functions and higher levels of compatibility with other technologies, such as Instant Message (Liu, 2016), instructional games (Rachels &

Rockinson-Szapkiw, 2018), and GPS (Chang et al., 2018). The finding that state-of-the-art technologies such as social networks and smartphones have superseded previously popular technologies such as chats and iPods indicates the development of TELL, as accompanied with the rapid improvements of mobile devices and Web 2.0 technologies.

Digital language learning games and AR are two typical technologies that are newly emerged and well developed in recent years, as found in our review and Shadiev and Yang (2020). Digital-game-based learning is one of the five most important types of TELL, the effects of which is generally positive, especially for enhancing vocabulary learning and improving learning experience (e.g. Castañeda & Cho, 2016; Hsu, 2017; Yukselturk et al., 2018). However, most publications on digital-game-based language learning before 2014 did not present explicit supporting data when discussing the effectiveness of the games (Golonka et al., 2014). We also found that AR technology has been increasingly employed for language learning in recent three years and has overall positive effects on knowledge acquisition and learner motivation (e.g. Ho et al., 2017; Hsu, 2017), although Golonka et al. (2014) did not report positive results concerning the effectiveness of AR. Such conflicting research results may stem from three main reasons. Firstly, the digital games and AR technology were not mature enough five years ago, and they were not well integrated in language learning to assist students to enter flow states (e.g. in deHaan, Reed, & Kuwada's (2010) research); however, they have been largely improved in recent years and can be well embedded in learning systems to increase the likelihood of causing flow states in students learning with the system (e.g. Hsu, 2017; Yukselturk et al., 2018). Secondly, previous researchers tended to conduct game-based learning and AR-featured learning in isolation, while recent researchers applied blended use of games and AR with other learning strategies, which could alleviate possible problems and realise full potential of the new technologies. For instance, Ho et al. (2017) pointed out that learner diversities from the perspective of cognitive styles were seldom considered in previous AR-featured u-learning, which may place negative influence on the learning outcomes. To solve this problem, they developed the instruction of AR-enhanced u-learning based on the theory of field dependence and field independence, and positive effects of this system in promoting learning were identified. Thirdly, there was 'lack of control groups using traditional methods' in previous TELL studies, which 'ma[d]e it impossible to evaluate whether learning was improved, impaired, or unaffected by these technologies' (Golonka et al., 2014, p. 82). However, studies in recent three years conducted comparisons between experimental groups and control groups (e.g. Ho et al.,

2017; Huang et al., 2016; Liakin et al., 2017), thus empirical data concerning the effectiveness of technologies can be collected and analysed.

In sum, future research on TELL may further explore the use of mobile technologies, social networks, digital games, and AR technology in language education, as they are the state-of-the-art technologies with great potential for effective learning and teaching.

#### ***4.3. Factors that may influence the effectiveness of TELL***

Not all studies reported the positive effects of TELL, although most did. Our results showed several factors that may influence the effectiveness of TELL, the first of which is the lack of training sessions that aimed to increase students' technology acceptance. Because students usually have years of experience learning languages under traditional approaches and have developed associated habits and strategies that are not necessarily useful for TELL (Lin, 2017), some of them may have felt unprepared for the change of learning mode, which might have negatively influenced their perceptions and learning motivation and thus the outcomes. We therefore suggest that in the classrooms typical of the studies reported here, training sessions be conducted to help students understand why e-learning is being applied, how it can assist their learning, what digital tools and devices are used, and how to use them to effectively participate in e-learning activities (Zou, 2020). Shyr and Chen (2018) demonstrated a good example of this when they trained their students before conducting a flipped classroom to familiarise them with the technologies used, which helped them adjust to the new pedagogy. The research results showed the positive effects of the flipped approach on learning achievements and motivation.

Second, the positive effects of TELL may decrease over time when learners lose the sense of freshness about the newly introduced technologies and become disengaged in the learning activities, as found by Wang et al. (2018). To address this problem, we suggest that teachers employ diversified modes of effective and innovative teaching strategies and skills and integrate them into their technology-enhanced teaching. Many studies have reported the positive effects of adding or applying new digital elements, new technology-enhanced strategies, or new classroom interaction modes to promote students' learning performance and motivation (Yang, Chang, Hwang, & Zou, 2020). Example of this include: the application of online peer assessment (Hsia, Huang, & Hwang, 2016); the use of a multi-level concept mapping-based question-posing approach (Hwang, Zou, & Lin, 2020), integration of technology-enhanced just-in-time teaching and peer instruction in flipped classrooms (Zou & Xie,

2019), and the implementation of game-based learning (Hung et al., 2018; Yukselturk et al., 2018). Diversified teaching approaches, modes, and strategies are essential for the maintenance of students' learning motivation and engagement.

Third, students' language proficiency may influence the extent to which they benefit from TELL, controversial results of which have appeared in the literature. On the one hand, Park and Warschauer (2016) and Shadiey, Huang, et al. (2017) identified statistically significant positive effects of TELL in assisting low-achievers but not higher-achievers. This is perhaps because low-achievers needed more help with information processing and comprehension; in addition, certain types of technology made the content more comprehensible and easier to process (Chen, Wang, Zou, Lin, et al., 2019). On the other hand, Song et al. (2017) found that TELL benefitted higher-achievers more as it might have increased the cognitive load on low-achievers, who may have felt certain types of technology too challenging. Such results indicate the importance of designing TELL activities and materials that are suitable for learners' language proficiency levels, as well as the necessity of providing low-achievers with more scaffolding and high-achievers with more challenges. Therefore, the development of more personalised learning systems appears to be an important task for future educators and researchers (Xie, Zou, Zhang, Wang, & Kwan, 2019; Zou & Xie, 2018).

#### **4.4. Limitations and future studies**

This research is limited in several aspects. Firstly, as we only reviewed original research articles that are published in 10 SSCI journals, the findings are preliminary. Clearly, if more publications from a wider range of sources (e.g. book chapters, conference papers, articles in journals that are indexed by AHCI, Scopus and ERIC, etc.) were included, the findings would be more comprehensive. Future research may work in this direction for results that can be better generalised.

Our list of papers for review was limited and some important studies on TELL were not included; thus, the technologies that we analysed did not cover all possibilities, for example, online or digital dictionaries, robots, virtual reality, and virtual environments. This is because we aimed at an in-depth review of high-quality empirical studies and applied very strict article selection criteria. Our search in the database turned up 220 articles, among which, six were on virtual reality and two on robots, but we excluded them from the review list either because they investigated first language learning, or because they did not examine the effectiveness of TELL. This leaves open the possibility for a more comprehensive review

that includes a wider range of technologies in the future. Nevertheless, our aim was not to summarise all types of technology but to analyse representative technologies in depth.

Additionally, we investigated the types, purposes, and effectiveness of the state-of-the-art technologies in second or foreign learning; however, we did not investigate the possible connection between the technologies and various aspects of language learning outcomes. Future studies may focus on this direction and investigate the optimal types and use of technologies in developing different aspects of language learning.

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## Appendix B. Technologies for the four main purposes

**Table B1.** Technologies for promoting practices.

| Aspects of learning outcomes | Effectiveness of state-of-the-art technologies                                                                                                                   | Types of state-of-the-art technologies for L2 learning                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Speaking proficiency         | Development of speaking proficiency was enhanced (Tecedor & Campos-Dintrans, 2019; Kurt & Bensen, 2017; Huang et al., 2016; Liu et al., 2016; Chen et al., 2016) | <ul style="list-style-type: none"> <li>• Interactive multimedia learning software (<i>Moodle</i>)</li> <li>• Mobile-based multimedia applications (<i>Story and Painting House, Vine</i>)</li> <li>• Multimedia technology (online voice recording, videoconferencing)</li> <li>• Speech-to-text recognition</li> <li>• Mobile technology (digital pen)</li> <li>• Automatic speech recognition technology</li> </ul> |
|                              | No significant effect on learners' speaking development was identified (Bodnar et al., 2017)                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                       |

(continued)

Table B1. Continued.

| Aspects of learning outcomes  | Effectiveness of state-of-the-art technologies                                                                                                                                                                            | Types of state-of-the-art technologies for L2 learning                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Listening proficiency         | Development of listening proficiency was enhanced (Tan et al., 2019; Liu, Chen, et al., 2018; Ho et al., 2017)                                                                                                            | <ul style="list-style-type: none"> <li>• Interactive multimedia learning software (<i>Livescribe</i>)</li> <li>• Multimedia technology</li> <li>• QR codes</li> <li>• Mobile technology (digital pen, smartphone)</li> <li>• Augmented Reality</li> <li>• Wiki (Wikitude World Browser)</li> </ul>                                                                                                                                                  |
| Reading proficiency           | Development of reading proficiency was enhanced (Darling-Aduana & Heinrich, 2018; Wang, 2017)                                                                                                                             | <ul style="list-style-type: none"> <li>• Multimedia technology</li> <li>• Mobile-based multimedia application (<i>Learn English Audio and Video</i>)</li> </ul>                                                                                                                                                                                                                                                                                     |
| Writing proficiency           | Development of writing proficiency was enhanced (Chen, Carger, et al., 2017; Talaván et al., 2017; Bikowski & Vithanage, 2016)                                                                                            | <ul style="list-style-type: none"> <li>• Mobile-based electronic book</li> <li>• Mobile-based digital handwriting application (<i>Penultimate</i>)</li> <li>• Wiki (Google Doc)</li> <li>• Multimedia technology</li> <li>• Subtitle making software (<i>Aegisub</i>)</li> </ul>                                                                                                                                                                    |
|                               | Learners' comprehension of essay writing was enhanced but their writing proficiency was not enhanced in the short term (Saricaoglu, 2019)                                                                                 | Computer-based automated formative feedback                                                                                                                                                                                                                                                                                                                                                                                                         |
| Vocabulary knowledge          | Development of vocabulary knowledge was enhanced (Sharifi, Soleimani, & Jafarigohar, 2017; Kurt & Bensen, 2017; Ho et al., 2017; Hsu, 2017; Hsieh et al., 2017; Marefat & Hassanzadeh, 2016; Liu, 2016; Liu et al., 2016) | <ul style="list-style-type: none"> <li>• E-portfolios</li> <li>• Mobile-based multimedia application (<i>Vine, Story and Painting House</i>)</li> <li>• Augmented Reality</li> <li>• Wiki (Wikitude World Browser)</li> <li>• Gamified learning</li> <li>• Social media (<i>Line</i>)</li> <li>• Mobile-assisted learning (Instant Message)</li> <li>• Multimedia technology</li> <li>• Mobile-based gamified learning (<i>Duolingo</i>)</li> </ul> |
|                               | No significant effect was found on learners' vocabulary development (Rachels & Rockinson-Szapkiw, 2018)                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Pronunciation                 | Development of pronunciation was enhanced (Qian, Chukharev-Hudilainen, & Levis, 2018)                                                                                                                                     | <ul style="list-style-type: none"> <li>• Corpus-based word frequency lists</li> <li>• Text-to-speech recognition</li> </ul>                                                                                                                                                                                                                                                                                                                         |
|                               | No immediate effect on learners' pronunciation was identified, but long-term effects were identified (Liakin et al., 2017)                                                                                                | <ul style="list-style-type: none"> <li>• Text-to-speech recognition</li> <li>• Interactive multimedia learning software (<i>Moodle</i>)</li> </ul>                                                                                                                                                                                                                                                                                                  |
| Grammar knowledge             | Development of grammar knowledge was enhanced (Talaván et al., 2017; Cerezo, 2016; Castañeda & Cho, 2016)                                                                                                                 | <ul style="list-style-type: none"> <li>• Multimedia technology</li> <li>• Subtitle making software (<i>Aegisub</i>)</li> <li>• Mobile-based gamified learning</li> </ul>                                                                                                                                                                                                                                                                            |
|                               | No significant effect on learners' grammar knowledge development was identified (Rachels & Rockinson-Szapkiw, 2018)                                                                                                       | <ul style="list-style-type: none"> <li>• Mobile-based gamified learning (<i>Duolingo</i>)</li> </ul>                                                                                                                                                                                                                                                                                                                                                |
| Comprehensive language skills | Development of comprehensive language skills was enhanced (Yukselturk et al., 2018; Shadiev, Hwang, Huang, et al., 2018; Shadiev, Huang, et al., 2017; Shadiev, Wu, et al., 2017; Hung, 2017; Yang, 2016)                 | <ul style="list-style-type: none"> <li>• Gamified learning (<i>Kahoot!</i>)</li> <li>• Kinect camera</li> <li>• Mobile-based electronic book</li> <li>• Multimedia technology</li> <li>• Social media (<i>Facebook, Skype</i>)</li> <li>• Clicker</li> <li>• Online learning platform (<i>ESP Café</i>)</li> <li>• Computer-aided translation</li> <li>• Speech-to-text recognition</li> </ul>                                                      |

(continued)

**Table B1.** Continued.

| Aspects of learning outcomes     | Effectiveness of state-of-the-art technologies                                                                                                                                                                                                                                                                                                                         | Types of state-of-the-art technologies for L2 learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Learning motivation and attitude | Learning motivation and attitude were enhanced (Tan et al., 2019; Shadiev, Hwang, Huang, et al., 2018; Liu, Chen, et al., 2018; Sharifi et al., 2017; Kurt & Bensen, 2017; Wang, 2017; Hsu, 2017; Chen, Carger, et al., 2017; Hung, 2017; Rachels & Rockinson-Szapkiw, 2018; Hsieh et al., 2017; Bikowski & Vithanage, 2016; Chen et al., 2016; Castañeda & Cho, 2016) | <ul style="list-style-type: none"> <li>• Interactive multimedia learning software (<i>Livescribe, Moodle</i>)</li> <li>• Mobile technology (digital pen, smartphone)</li> <li>• Mobile-based electronic book</li> <li>• Multimedia technology</li> <li>• QR codes</li> <li>• E-portfolios</li> <li>• Clicker</li> <li>• Mobile-based multimedia applications (<i>Learn English Audio and Video, Vine</i>)</li> <li>• Gamified learning (<i>Kahoot!, Duolingo</i>)</li> <li>• Mobile-based digital handwriting application (<i>Penultimate</i>)</li> <li>• Augmented Reality</li> <li>• Social media (<i>Line</i>)</li> <li>• Wiki (<i>Google Doc</i>)</li> <li>• Computer-generated corrective feedback to learners' practices</li> </ul> |
| Learning attention               | <p>Learning motivation was reduced (Cerezo, 2016)</p> <p>Learning attention was enhanced (Shadiev, Huang, et al., 2017; Shadiev, Wu, et al., 2017)</p>                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Social media (<i>Facebook, Skype</i>)</li> <li>• Speech-to-text recognition</li> <li>• Computer-aided translation</li> <li>• Multimedia technology</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

**Table B2.** Technologies for delivering instructional content.

| Aspects of learning outcomes | Effectiveness of state-of-the-art technologies                                                                                                                                                                                                                                                                                                                                                                                              | Types of state-of-the-art technologies for L2 learning                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Speaking proficiency         | <p>Development of speaking proficiency was enhanced (Liu, Tai, et al., 2018; Lin &amp; Hwang, 2018; Huang et al., 2016; Lan &amp; Lin, 2016; Hwang et al., 2016)</p> <p>No significant effect on learners' speaking development was identified (Bodnar et al., 2017)</p> <p>Development of speaking fluency was enhanced.</p> <p>No significant effect on learners' speaking complexity and accuracy was identified (Wang et al., 2018)</p> | <ul style="list-style-type: none"> <li>• Mobile-based multimedia applications (<i>Story and Painting House, MOBILE SEAMLESS</i>)</li> <li>• Social media (<i>Facebook</i>)</li> <li>• Multimedia technology</li> <li>• Speech-to-text recognition</li> <li>• Web-based multimedia storytelling</li> <li>• Automatic speech recognition technology</li> <li>• MOOC</li> </ul>                                                                                                                |
| Listening proficiency        | Development of listening proficiency was enhanced (Liu, Chen, et al., 2018; Chang et al., 2018)                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>• Multimedia technology</li> <li>• Mobile technology (smartphones)</li> <li>• QR codes</li> <li>• Fitness machines</li> <li>• GPS</li> <li>• Mobile-based multimedia applications (<i>Noteledge, Learn English Audio and Video</i>)</li> <li>• Multimedia technology</li> <li>• Mobile-based electronic book</li> <li>• Multimedia technology</li> <li>• Visual-syntactic text formatting technology for visualising syntactic structures</li> </ul> |
| Reading proficiency          | <p>Development of reading proficiency was enhanced (Liu, Huang, et al., 2018; Wang, 2017)</p> <p>No significant effect on learners' reading was identified (Lin, 2017; Park &amp; Warschauer, 2016)</p>                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>• Multimedia technology</li> <li>• Mobile-based electronic book</li> <li>• Multimedia technology</li> <li>• Visual-syntactic text formatting technology for visualising syntactic structures</li> </ul>                                                                                                                                                                                                                                              |

(continued)

Table B2. Continued.

| Aspects of learning outcomes     | Effectiveness of state-of-the-art technologies                                                                                                                                                                                                                             | Types of state-of-the-art technologies for L2 learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Writing proficiency              | Development of writing proficiency was enhanced (Tsai, 2019)<br>No significant effect on high achievers' L2 writing was identified, but the low achievers' development of written conventions and writing strategies was enhanced (Park & Warschauer, 2016)                | <ul style="list-style-type: none"> <li>• Multimedia technology</li> <li>• Interactive courseware</li> <li>• Multimedia technology</li> <li>• Visual-syntactic text formatting technology for visualising syntactic structures</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Vocabulary knowledge             | Development of vocabulary knowledge was enhanced (Connor et al., 2019; Liu, Huang, et al., 2018; Marefat & Hassanzadeh, 2016; Liu, 2016)<br><br>No significant effect on learners' vocabulary knowledge development was identified (Rachels & Rockinson-Szapkiw, 2018)     | <ul style="list-style-type: none"> <li>• Mobile-based electronic book</li> <li>• Mobile-based multimedia application (<i>Noteledge</i>)</li> <li>• Multimedia technology</li> <li>• Mobile-assisted learning (Instant Message)</li> <li>• Mobile-based gamified learning (<i>Duolingo</i>)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                     |
| Pronunciation                    | Development of pronunciation was enhanced (Qian et al., 2018; Okuno & Hardison, 2016)<br>No immediate effect on learners' pronunciation was identified, but the long-term effect was found (Liakin et al., 2017)                                                           | <ul style="list-style-type: none"> <li>• Corpus-based word frequency lists</li> <li>• Text-to-speech recognition</li> <li>• Multimedia technology</li> <li>• Text-to-speech recognition</li> <li>• Interactive multimedia learning software (<i>Moodle</i>)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                    |
| Grammar knowledge                | Development of grammar knowledge was enhanced (Castañeda & Cho, 2016)                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>• Mobile-based gamified learning</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Comprehensive language skills    | Development of comprehensive language skills was enhanced (Wang et al., 2019; Yukselturk et al., 2018; Shadiev, Hwang, Huang, et al., 2018; Tseng et al., 2018; Shadiev, Huang, et al., 2017; Yang, 2016)                                                                  | <ul style="list-style-type: none"> <li>• QR codes</li> <li>• Mobile technology (smartphone)</li> <li>• Gamified learning</li> <li>• Kinect camera</li> <li>• Multimedia technology</li> <li>• Mobile-based electronic book</li> <li>• Online learning platforms (<i>Zaption</i>, <i>Quizlet</i>, <i>ESP Café</i>)</li> <li>• Mobile-based multimedia applications (<i>Padlet</i>, <i>Flipgrid</i>)</li> <li>• Social media (<i>Facebook</i>, <i>Skype</i>)</li> <li>• Speech-to-text recognition</li> <li>• Computer-aided translation</li> </ul>                                                                                                                         |
| Learning motivation and attitude | Learning motivation and attitude were enhanced (Wang et al., 2019; Tsai, 2019; Shadiev, Hwang, Huang, et al., 2018; Tseng et al., 2018; Rachels & Rockinson-Szapkiw, 2018; Shadiev, Huang, et al., 2017; Wang, 2017; Lin, 2017; Hwang et al., 2016; Castañeda & Cho, 2016) | <ul style="list-style-type: none"> <li>• QR codes</li> <li>• Mobile technology (smartphone)</li> <li>• Multimedia technology</li> <li>• Mobile-based electronic book</li> <li>• Interactive courseware</li> <li>• Online learning platforms (<i>Zaption</i>, <i>Quizlet</i>)</li> <li>• Mobile-based multimedia applications (<i>Padlet</i>, <i>Flipgrid</i>, <i>Learn English Audio and Video</i>, <i>GBELA</i>)</li> <li>• Mobile-based gamified learning (<i>Duolingo</i>)</li> <li>• Social media (<i>Facebook</i>, <i>Skype</i>)</li> <li>• Speech-to-text recognition</li> <li>• Computer-aided translation</li> <li>• Web-based multimedia storytelling</li> </ul> |
| Cognitive load                   | Cognitive load was reduced (Chang et al., 2018; Huang et al., 2016)<br><br>Peer-to-peer collaboration and interactions were enhanced (Lan & Lin, 2016)                                                                                                                     | <ul style="list-style-type: none"> <li>• GPS</li> <li>• Mobile technology (smartphone)</li> <li>• Speech-to-text recognition</li> <li>• Mobile-based multimedia application (<i>MOBILE SEamless</i>)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |



**Table B3.** Technologies for facilitating interactions.

| Aspects of learning outcomes  | Effectiveness of state-of-the-art technologies                                                                                                                                                   | Types of state-of-the-art technologies for L2 learning                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Speaking proficiency          | Development of speaking proficiency was enhanced (Tecedor & Campos-Dintrans, 2019; Yeh & Lai, 2019; Nguyen et al., 2018; Lin & Hwang, 2018; Kurt & Bensen, 2017; Alvarez-Marinelli et al., 2016) | <ul style="list-style-type: none"> <li>• Multimedia technology (online voice recording, videoconferencing)</li> <li>• Social media (<i>Skype, Facebook</i>)</li> <li>• Mobile-based multimedia applications (<i>ezTranslate, Vine</i>)</li> <li>• Speech-to-text recognition</li> <li>• Computer-assisted language learning (Picture Vocabulary, Verbal Analogies, Understanding Directions, Story Recall)</li> </ul> |
| Listening proficiency         | Development of listening proficiency was enhanced (Nguyen et al., 2018; Chang et al., 2018; Liu, Chen, et al., 2018; Alvarez-Marinelli et al., 2016)                                             | <ul style="list-style-type: none"> <li>• Mobile-based multimedia application (<i>ezTranslate</i>)</li> <li>• Speech-to-text recognition</li> <li>• GPS</li> <li>• Mobile technology (smartphone)</li> <li>• Multimedia technology</li> <li>• QR codes</li> <li>• Computer-assisted language learning (Picture Vocabulary, Verbal Analogies, Understanding Directions, Story Recall)</li> </ul>                        |
| Reading proficiency           | Development of reading proficiency was enhanced (Darling-Aduana & Heinrich, 2018; Wang, 2017)                                                                                                    | <ul style="list-style-type: none"> <li>• Mobile-based electronic book</li> <li>• Multimedia technology</li> <li>• Mobile-based multimedia application (<i>Learn English Audio and Video</i>)</li> </ul>                                                                                                                                                                                                               |
| Writing proficiency           | Development of writing proficiency was enhanced (Lin et al., 2018; Chen, Carger, et al., 2017; Talaván et al., 2017; Bikowski & Vithanage, 2016)                                                 | <ul style="list-style-type: none"> <li>• Gamified learning</li> <li>• Mobile-based digital handwriting application (<i>Penultimate</i>)</li> <li>• Multimedia technology</li> <li>• Subtitle making software (<i>Aegisub</i>)</li> <li>• Wiki (Google Doc)</li> <li>• Social media (<i>Facebook, WhatsApp, Vine, Line</i>)</li> </ul>                                                                                 |
| Vocabulary knowledge          | Development of vocabulary knowledge was enhanced (Çetinkaya & Sütçü, 2018; Kurt & Bensen, 2017; Hsieh et al., 2017; Alvarez-Marinelli et al., 2016)                                              | <ul style="list-style-type: none"> <li>• Computer-assisted language learning (Picture Vocabulary, Verbal Analogies, Understanding Directions, Story Recall)</li> </ul>                                                                                                                                                                                                                                                |
| Pronunciation                 | Development of pronunciation was enhanced (Roussel & Galan, 2018)                                                                                                                                | <ul style="list-style-type: none"> <li>• Clicker</li> </ul>                                                                                                                                                                                                                                                                                                                                                           |
| Grammar knowledge             | Development of grammar knowledge was enhanced (Talaván et al., 2017)                                                                                                                             | <ul style="list-style-type: none"> <li>• Multimedia technology</li> <li>• Subtitle making software (<i>Aegisub</i>)</li> </ul>                                                                                                                                                                                                                                                                                        |
| Comprehensive language skills | Development of comprehensive language skills was enhanced (Shadiev, Hwang, Huang, et al., 2018; Tseng et al., 2018; Hung, 2017)                                                                  | <ul style="list-style-type: none"> <li>• Mobile-based electronic book</li> <li>• Multimedia technology</li> <li>• Online learning platforms (<i>Zaption, Quizlet</i>)</li> <li>• Mobile-based multimedia applications (<i>Padlet, Flipgrid</i>)</li> <li>• Clicker</li> <li>• Gamified learning (<i>Kahoot!</i>)</li> </ul>                                                                                           |

(continued)

**Table B3.** Continued.

| Aspects of learning outcomes     | Effectiveness of state-of-the-art technologies                                                                                                                                                                                                                                                                                                                    | Types of state-of-the-art technologies for L2 learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Learning motivation and attitude | Learning motivation and attitude were enhanced (Nguyen et al., 2018; Lin et al., 2018; Çetinkaya & Sütçü, 2018; Chang et al., 2018; Shadiëv, Hwang, Huang, et al., 2018; Lin & Hwang, 2018; Tseng et al., 2018; Liu, Chen, et al., 2018; Kurt & Bensen, 2017; Wang, 2017; Chen, Carger, et al., 2017; Hsieh et al., 2017; Hung, 2017; Bikowski & Vithanage, 2016) | <ul style="list-style-type: none"> <li>• Mobile-based multimedia applications (<i>ezTranslate, Padlet, Flipgrid, Vine, Learn English Audio and Video</i>)</li> <li>• Speech-to-text recognition</li> <li>• Gamified learning (<i>Kahoot!</i>)</li> <li>• Social media (<i>Facebook, WhatsApp, Line</i>)</li> <li>• GPS</li> <li>• Mobile technology (smartphone)</li> <li>• Mobile-based electronic book</li> <li>• Multimedia technology</li> <li>• Online learning platforms (<i>Zaption, Quizlet</i>)</li> <li>• QR codes</li> <li>• Mobile-based digital handwriting application (<i>Penultimate</i>)</li> <li>• Clicker</li> <li>• Wiki (<i>Google Doc</i>)</li> <li>• Clicker</li> <li>• GPS</li> <li>• Mobile technology (smartphone)</li> </ul> |
| Cognitive load                   | Cognitive load was reduced (Roussel & Galan, 2018; Chang et al., 2018)                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>• Customised personal response system for students' questioning and answering and for instructors' feedback provision</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Learning attention               | Learning attention was enhanced (Song et al., 2017)                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>• Social media (<i>Facebook</i>)</li> <li>• Multimedia technology</li> <li>• Customised personal response system for students' questioning and answering and for instructors' feedback provision</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Interaction                      | Peer-to-peer collaboration and interactions were enhanced (Lin & Hwang, 2018; Song et al., 2017)                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>• Customised personal response system for students' questioning and answering and for instructors' feedback provision</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

**Table B4.** Technologies for restructuring teaching approaches.

| Aspects of learning outcomes     | Effectiveness of state-of-the-art technologies                                                                                                                                                                                                                                                                                                               | Types of state-of-the-art technologies for L2 learning                                                                                                                                                                                                                                                              |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Speaking proficiency             | <p>Development of speaking proficiency was enhanced (Tecedor &amp; Campos-Dintrans, 2019; Yeh &amp; Lai, 2019; Lin &amp; Hwang, 2018; Kurt &amp; Bensen, 2017; Wu, Hsieh, &amp; Yang, 2017).</p> <p>Development of speaking fluency was enhanced. No significant effect on learners' speaking complexity and accuracy was identified (Wang et al., 2018)</p> | <ul style="list-style-type: none"> <li>• Multimedia technology (online voice recording, videoconferencing)</li> <li>• Social media (<i>Skype</i>, <i>Facebook</i>, <i>Line</i>)</li> <li>• Mobile-based multimedia application (<i>Vine</i>)</li> <li>• MOOC</li> </ul>                                             |
| Reading proficiency              | Development of reading proficiency was enhanced (Huang & Chuang, 2016)                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>• Interactive multimedia learning software (PowerCam software)</li> <li>• Multimedia technology</li> <li>• Online learning platform</li> <li>• Gamified learning</li> </ul>                                                                                                  |
| Writing proficiency              | Development of writing proficiency was enhanced (Lin et al., 2018)                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                     |
| Vocabulary knowledge             | Development of vocabulary knowledge was enhanced (Hsieh et al., 2017; Kurt & Bensen, 2017; Marefat & Hassanzadeh, 2016)                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>• Social media (<i>Line</i>)</li> <li>• Mobile-based multimedia application (<i>Vine</i>)</li> <li>• Multimedia technology</li> </ul>                                                                                                                                        |
| Comprehensive language skills    | Development of comprehensive language skills was enhanced (Haghighi et al., 2019; Shyr & Chen, 2018; Hung, 2017; Yang, 2016)                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Social media (<i>Telegram</i>)</li> <li>• Online learning platforms (<i>Flip2Learn</i>, <i>ESP Café</i>)</li> <li>• Gamified learning (<i>Kahoot!</i>)</li> <li>• Multimedia technology</li> <li>• Social media (<i>Telegram</i>, <i>Facebook</i>, <i>Line</i>)</li> </ul> |
| Learning motivation and attitude | Learning motivation and attitudes were enhanced (Haghighi et al., 2019; Lin et al., 2018; Lin & Hwang, 2018; Hsieh et al., 2017; Kurt & Bensen, 2017; Hung, 2017; Huang & Chuang, 2016)                                                                                                                                                                      | <ul style="list-style-type: none"> <li>• Gamified learning</li> <li>• Multimedia technology</li> <li>• Mobile-based multimedia application (<i>Vine</i>)</li> <li>• Gamified learning (<i>Kahoot!</i>)</li> <li>• Interactive multimedia learning software (PowerCam software)</li> </ul>                           |
| Learning attention               | Learning attention was enhanced (Huang & Chuang, 2016)                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>• Online learning platform</li> <li>• Interactive multimedia learning software (PowerCam software)</li> <li>• Multimedia technology</li> <li>• Online learning platform</li> </ul>                                                                                           |
| Interaction                      | Peer-to-peer collaborations and interactions were enhanced (Lin & Hwang, 2018; Wu et al., 2017)                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• Social media (<i>Facebook</i>, <i>Line</i>)</li> <li>• Multimedia technology</li> </ul>                                                                                                                                                                                    |

## Appendix C. Types, purposes, and effectiveness of state-of-the-art technologies for L2 learning

| Author                          | Types of state-of-the-art technologies for L2 learning                                                                | Purposes of state-of-the-art technologies in L2 learning                                                                    | Effectiveness of state-of-the-art technologies in L2 learning                                                                 |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Alvarez-Marinelli et al. (2016) | Computer-assisted language learning, Picture Vocabulary, Verbal Analogies, Understanding Directions, and Story Recall | Facilitating peer-to-peer and tutor-student communication in class                                                          | Development of speaking and listening proficiency and vocabulary knowledge was enhanced.                                      |
| Bikowski and Vithanage (2016)   | Google Doc                                                                                                            | Assisting in-class collaborative writing                                                                                    | Development of writing proficiency and learning attitudes were enhanced.                                                      |
| Bodnar et al. (2017)            | Automatic speech recognition technology                                                                               | Automatically recognising and pointing out errors in learners' speeches                                                     | No significant effect on learners' speaking was identified.                                                                   |
| Castañeda and Cho (2016)        | Mobile-based digital game                                                                                             | Encouraging learners' practices of grammar and promoting learning engagement                                                | Development of grammar knowledge and learning confidence were enhanced.                                                       |
| Cerezo (2016)                   | Computer-generated corrective feedback to learners' practices                                                         | Providing immediate evaluation of learners' output, feedback to errors, and personalised practice contents                  | Development of grammar knowledge was enhanced, but the learning motivation was reduced.                                       |
| Çetinkaya and Sütçü (2018)      | WhatsApp, Facebook                                                                                                    | Serving as a peer-to-peer interaction platform for learners' vocabulary learning                                            | Development of word knowledge and learning attitude were enhanced. WhatsApp was more effective than Facebook.                 |
| Chang et al. (2018)             | Combination of ubiquitous outdoor learning and indoor computer-assisted learning by GPS and mobile devices            | Contextualising the instructional contents and providing instructions anywhere anytime                                      | Development of listening comprehension and learning motivation were enhanced. Cognitive load in learning process was reduced. |
| Chen, Carger, et al. (2017)     | <i>Penultimate</i> , an iPad-based commercial app for digital handwriting                                             | Facilitating the in-class narrative writing                                                                                 | Development of writing proficiency and learning attitude were enhanced.                                                       |
| Chen et al. (2016)              | Digital pen, Moodle                                                                                                   | Expediting the repeated reading strategy for promoting oral reading fluency, learning motivation, and learning satisfaction | Development of oral reading fluency, learning motivation, and learning satisfaction were enhanced.                            |
| Connor et al. (2019)            | Tablet-based multimedia e-book embedded with comprehension questions                                                  | Monitoring learners' comprehension of the instructional contents and providing personalised learning experiences            | Development of word knowledge and learning strategies was enhanced.                                                           |

(continued)

## Appendix C. Continued.

| Author                             | Types of state-of-the-art technologies for L2 learning                        | Purposes of state-of-the-art technologies in L2 learning                                                                                                          | Effectiveness of state-of-the-art technologies in L2 learning                                                                                                                                         |
|------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Darling-Aduana and Heinrich (2018) | Tablet (eReader)-based multimedia learning with accessibility to the Internet | Assisting the low-income English learners in both English/Spanish bilingual and traditional English-only reading classrooms                                       | Development of reading proficiency was enhanced better in the bilingual classroom than in the traditional classroom.                                                                                  |
| Haghighi et al. (2019)             | <i>Telegram</i>                                                               | Supporting the flipped classroom by posting course materials through Telegram before class, and conducting communicative activities in class                      | Development of pragmatic competence and learning satisfaction were enhanced.                                                                                                                          |
| Ho et al. (2017)                   | Augmented Reality , Wikitude World Browser                                    | Facilitating learners' English listening and speaking practices by providing contextualised digital information and artificial learning contexts anywhere anytime | Individuals with field-dependent cognitive style performed better than those with field independent and mix field cognitive styles in terms of the development of speaking and listening proficiency. |
| Hsieh et al. (2017)                | <i>Line</i>                                                                   | Assisting students of a flipped oral training course in practicing target English idioms beyond the classroom through written and verbal discussions              | Development of idiomatic knowledge and learning motivation were enhanced.                                                                                                                             |
| Hsu (2017)                         | Digital game enhanced by Augmented Reality technologies                       | Encouraging vocabulary learning and practicing within artificial learning contexts                                                                                | Development of word knowledge, learning motivation, and learning experiences were enhanced.                                                                                                           |
| Huang and Chuang (2016)            | E-learning course platform, PowerCam software, video                          | Supporting a content-based reading course where the language learning focused more on meaning and content than form                                               | Development of reading comprehension, reading speed, learning engagement, and learning attitude were enhanced.                                                                                        |
| Huang et al. (2016)                | Speech-to-text recognition system                                             | Reducing learners' cognitive load                                                                                                                                 | Development of speaking proficiency was enhanced, and low achievers' cognitive load was reduced.                                                                                                      |
| Hung (2017)                        | Clicker, <i>Kahoot!</i> game                                                  | Facilitating students' question-and-answer in class and promoting in-class interactions                                                                           | Development of comprehensive language skills and learning motivation were enhanced.                                                                                                                   |
| Hwang et al. (2016)                | Web-based multimedia storytelling                                             | Supporting speaking classrooms by presenting instructional contents through multimedia                                                                            | Development of speaking proficiency and learning motivation were enhanced.                                                                                                                            |

(continued)

## Appendix C. Continued.

| Author                    | Types of state-of-the-art technologies for L2 learning                                   | Purposes of state-of-the-art technologies in L2 learning                                                                                                            | Effectiveness of state-of-the-art technologies in L2 learning                                                                                                     |
|---------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kurt and Bensen (2017)    | <i>Vine</i> , a mobile application that allowed the recording and sharing of video clips | Encouraging learners' speaking practices and word knowledge application                                                                                             | Development of vocabulary knowledge, speaking proficiency, and learning motivation were enhanced.                                                                 |
| Lan and Lin (2016)        | <i>MOBILE SEAMLESS</i> , a mobile application which created real-world contexts          | Supporting learners' task-induced language learning in real-world contexts                                                                                          | Development of pragmatic speaking proficiency and peer-to-peer collaborations were enhanced.                                                                      |
| Liakin et al. (2017)      | Text-to-speech recognition system, <i>Moodle</i>                                         | Assisting learners' practices in speaking and pronunciation                                                                                                         | No immediate effect on learners' pronunciation was identified, but the long-term effect was found.                                                                |
| Lin (2017)                | Tablet-based multimedia e-book                                                           | Delivering the textbook contents in the language classroom                                                                                                          | No significant effect on learners' reading comprehension was identified, but learning motivation was enhanced.                                                    |
| Lin and Hwang (2018)      | <i>Facebook</i> , videos                                                                 | Supporting the flipped language classroom by providing video instructions and encouraging speaking practices and peer-to-peer interactions in- and out-of-classroom | Development of pragmatic speaking proficiency, peer-to-peer interactions, and learning motivation were enhanced.                                                  |
| Lin et al. (2018)         | Contextual digital game                                                                  | Encouraging learners' knowledge application by providing interesting artificial environments                                                                        | Development of writing proficiency, learning experience, and learning interest were enhanced.                                                                     |
| Liu (2016)                | Instant Message function (IM) of mobile phones                                           | Facilitating learners' vocabulary learning by helping them exchange, receive, and create multimedia electronic word knowledge                                       | Development of vocabulary knowledge was enhanced.                                                                                                                 |
| Liu, Chen, et al. (2018)  | Video, context-aware tools that consisted of smartphones, QR codes and fitness machines  | Encouraging learners' contextualised practices of listening comprehension                                                                                           | Development of listening comprehension, listening strategies, and learning attitudes were enhanced                                                                |
| Liu, Huang, et al. (2018) | <i>Noteledge</i> , a tablet-based multimedia storytelling application                    | Supporting learners' individual and collaborative practices of story reading                                                                                        | Development of reading comprehension and vocabulary knowledge was enhanced. The students who learned collaboratively outperformed those who learned individually. |

(continued)

## Appendix C. Continued.

| Author                         | Types of state-of-the-art technologies for L2 learning                                                                             | Purposes of state-of-the-art technologies in L2 learning                                                                                                                                       | Effectiveness of state-of-the-art technologies in L2 learning                                                                                                                                          |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Liu, Tai, et al. (2018)        | <i>Story and Painting House</i> , a tablet-based multimedia storytelling application                                               | Supporting the free-space digital storytelling and encouraging learners' speaking practices                                                                                                    | Development of speaking accuracy, creativity, and learning motivation were enhanced.                                                                                                                   |
| Liu et al. (2016)              | <i>Story and Painting House</i> , a tablet-based multimedia storytelling application                                               | Supporting the free-space digital storytelling and encouraging learners' speaking practices                                                                                                    | Development of speaking fluency and vocabulary knowledge was enhanced. Learners' engagement was low at the beginning but increased later.                                                              |
| Marefat and Hassanzadeh (2016) | Video                                                                                                                              | Supporting the delivery of classroom-based form-focused instructions that drew learners' attention to the forms and structures of the language within the context of communicative interaction | Development of vocabulary knowledge was enhanced.                                                                                                                                                      |
| Nguyen et al. (2018)           | <i>ezTranslate</i> , a mobile application supporting the instant speech-to-text translation                                        | Facilitating the contextualised and ubiquitous practices of speaking and listening and encouraging peer-to-peer cooperative learning                                                           | Development of speaking and listening and learning motivation were enhanced.                                                                                                                           |
| Okuno and Hardison (2016)      | Multimedia technologies                                                                                                            | Facilitating learners' training and learning of pronunciation                                                                                                                                  | Both audio-visual and audio-only training helped enhance learners' pronunciation. Audio-visual training was more effective.                                                                            |
| Park and Warschauer (2016)     | Multimedia technologies, Visual-syntactic text formatting technology that visualised syntactic structures                          | Facilitating learners' reading and learning by converting the textbook to the one with syntactic enhancement                                                                                   | No significant effects on reading or on high-proficient learners' L2 writing were identified, but the low-proficient learners' development of written conventions and writing strategies was enhanced. |
| Qian et al. (2018)             | Computer-based segmental perceptual training system that utilised the corpus-based word frequency lists, text-to-speech technology | Personalising the phonetic input according to learners' individual learning perceptions                                                                                                        | Development of pronunciation was enhanced.                                                                                                                                                             |

(continued)



## Appendix C. Continued.

| Author                               | Types of state-of-the-art technologies for L2 learning                                                                   | Purposes of state-of-the-art technologies in L2 learning                                                                                                                       | Effectiveness of state-of-the-art technologies in L2 learning                                                                                                 |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rachels and Rockinson-Szapkiw (2018) | <i>Duolingo</i> , a mobile-based game application                                                                        | Encouraging learners' vocabulary and grammar learning and practices by providing personalised learning experiences based on students' personal preferences and learning styles | No significant effects on learners' vocabulary and grammar development were identified, but their learning motivation was enhanced.                           |
| Roussel and Galan (2018)             | Clicker                                                                                                                  | Expediting learners' self-reporting and opinion-voicing in vocabulary classrooms                                                                                               | Development of pronunciation was enhanced, and learners' cognitive load was alleviated.                                                                       |
| Saricaoglu (2019)                    | Computer-based automated formative feedback                                                                              | Facilitating learners' cause-and-effect essay writing                                                                                                                          | Learners' comprehension of essay writing was enhanced, but their writing proficiency was not enhanced in the short term.                                      |
| Shadiev, Hwang, Huang, et al. (2018) | Tablet-based electronic textbook, multimedia technologies                                                                | Supporting learners' textbook learning in individual and collaboration modes                                                                                                   | Learners in collaboration had their development of comprehensive language skills and learning motivation enhanced.                                            |
| Shadiev, Hwang, and Liu (2018)       | Multimedia technologies, smart-watch-based learning system that supported text-to-speech and speech-to-text technologies | Facilitating learners' writing by providing synchronous corrections and language support                                                                                       | Development of writing, grammar knowledge, and learning motivation were enhanced.                                                                             |
| Shadiev, Huang, et al. (2017)        | Facebook, Skype, speech-to-text recognition technology, computer-aided translation system                                | Supporting learners' online cross-culture communication and language learning                                                                                                  | Development of comprehensive language skills, learning attention, and mediation were enhanced. Female learners had higher levels of attention and meditation. |
| Shadiev, Wu, et al. (2017)           | Multimedia technologies, speech-to-text recognition technology                                                           | Expediting language learning during the lectures by providing synchronous language support                                                                                     | Development of comprehensive language skills, learning attention, and mediation were enhanced. In long-term, the learning attention declined.                 |
| Sharifi et al. (2017)                | E-portfolios                                                                                                             | Encouraging learning engagement by providing opportunities to apply pedagogical concepts                                                                                       | Development of vocabulary knowledge and learning motivation were enhanced.                                                                                    |

(continued)

## Appendix C. Continued.

| Author                             | Types of state-of-the-art technologies for L2 learning                                                                                   | Purposes of state-of-the-art technologies in L2 learning                                                                                    | Effectiveness of state-of-the-art technologies in L2 learning                                                                   |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Shyr and Chen (2018)               | <i>Flip2Learn</i> , an online learning platform                                                                                          | Supporting the flipped language classroom by facilitating the out-of-class learning section and encouraging teacher–student communication   | Development of comprehensive language skills was enhanced.                                                                      |
| Song et al. (2017)                 | Customised personal response system in which students could generate questions and answer questions for the instructors to give feedback | Expediting the question-and-answer section between instructors and students                                                                 | Learners' engagement with the learning and the in-class interactions were increased.                                            |
| Talaván et al. (2017)              | Video, <i>Aegisub</i> , a subtitle making software                                                                                       | Facilitating the learning of grammar and writing by supporting learners' collaborative reverse subtitling                                   | Development of grammar knowledge and writing proficiency was enhanced.                                                          |
| Tan et al. (2019)                  | <i>Livescribe</i> , a multimedia computing platform, interactive board, digital pens                                                     | Facilitating learners' practices of listening exercises                                                                                     | Development of listening comprehension and learning motivation were enhanced.                                                   |
| Tecedor and Campos-Dintrans (2019) | Online voice recording, videoconferencing                                                                                                | Facilitating learner's speaking practices by supporting the peer-to-peer computer-mediated communications                                   | Development of speaking complexity and fluency was enhanced.                                                                    |
| Tseng et al. (2018)                | <i>Padlet</i> , <i>Flipgrid</i> , <i>Zaption</i> , <i>Quizlet</i> , multimedia technologies                                              | Supporting the immersive flipped language classroom by providing instructional contents pre-class and encouraging out-of-class interactions | Development of comprehensive language skills and learning motivation were enhanced.                                             |
| Tsai (2019)                        | Multimedia technologies, Interactive courseware                                                                                          | Expediting learners' writing practices                                                                                                      | Development of writing proficiency and learning attitudes were enhanced.                                                        |
| Wang (2017)                        | <i>Learn English Audio and Video</i> , a mobile application for language learning, multimedia technologies                               | Supporting self-paced learning by providing multimedia language learning materials and exercises anywhere anytime                           | Development of reading proficiency and learning satisfaction were enhanced.                                                     |
| Wang et al. (2018)                 | MOOC                                                                                                                                     | Supporting the flipped language classroom by integrating several online instructional platforms and resources                               | Development of speech fluency was enhanced. No significant effect on learners' speaking complexity and accuracy was identified. |

(continued)

## Appendix C. Continued.

| Author                   | Types of state-of-the-art technologies for L2 learning                                                                | Purposes of state-of-the-art technologies in L2 learning                                                                                          | Effectiveness of state-of-the-art technologies in L2 learning                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Wang et al. (2019)       | <i>Green-building English-learning</i> , a mobile application, QR codes                                               | Providing contextualised instructional materials according to learners' surroundings where they scanned the QR codes                              | Development of comprehensive language skills and self-efficacy was enhanced for both high- and low-proficient learners. |
| Wu et al. (2017)         | Video, <i>Line</i>                                                                                                    | Supporting the flipped language classroom by providing instructional contents before class and encouraging peer-to-peer interactions out-of-class | Development of speaking proficiency and peer-to-peer cooperation was enhanced.                                          |
| Yang (2016)              | <i>ESP Café</i> , an online learning workshop                                                                         | Facilitating self-directed language learning by providing learning materials and encouraging discussions and practices                            | Development of comprehensive language skills was enhanced.                                                              |
| Yeh and Lai (2019)       | Skype                                                                                                                 | Supporting online communication between teachers and students                                                                                     | Development of speaking proficiency was enhanced, especially for the low-proficient learners.                           |
| Yukselturk et al. (2018) | Digital games, Kinect camera technology that gave synchronous commands depending on learners' responses during gaming | Facilitating self-directed learning in interesting learning contexts and providing immediate feedback                                             | Development of comprehensive language skills was enhanced.                                                              |